



Fall Semester 2025-2026

Digital Assignment-I

Course Name & code: Data Structures and Algorithms-BCSE202L

Program & Branch: B-Tech

Class Number: VL2025260101553

Marks: 10

Faculty Name: Dr.S.M Prabin

Last Date: 31-08-2025

**General instruction(s):**

1. Answer all questions

**Questions**

1. Solve the Following recurrences using Master theorem

1.  $T(n) = 3T\left(\frac{n}{2}\right) + \sqrt{n}$

2.  $T(n) = 4T(n/4) + n^2 \log n$

3.  $T(n) = 2T\left(\frac{n}{4}\right) + \frac{1}{n}$

4.  $T(n) = 16T(n/4) - n^2 \log n$

2. i) Find the time complexities of following recurrence relation using recursion tree method:

- a.  $T(n) = 4\left(\frac{n}{2}\right) + n$

- b.  $T(n) = T(n-2) + T(n-1) + n$

- ii) Solve the following recurrence using Substitution method.

$$T(n) = \begin{cases} 1, & \text{if } n = 1 \\ T(n-1) + n(n-1), & \text{if } n \geq 2 \end{cases}$$

$$T(n) = 2T(\sqrt{n}) + 1$$

3. Illustrate the conversion of the following infix expression  $b - c * a + (d * e - f) / g * (f^2)$  to postfix and prefix expression and give the algorithm.

4. Assume that the operation ENQUEUE (CQ, x) inserts an item x into a circular queue CQ and another operation DEQUEUE (CQ) deletes an item from CQ in FIFO manner. Draw the circular queue of size 7. Illustrate the working for the following ten commands (in the given order) over that circular queue (illustrate with diagram for execution of every command), also give the values of the variables FRONT and REAR on execution of each command;

- (i) ENQUEUE (CQ, 20); (ii) ENQUEUE (CQ, 40); (iii) ENQUEUE (CQ, 60); (iv) DEQUEUE (CQ); (v) ENQUEUE (CQ, 80); (vi) ENQUEUE (CQ, 100); (vii) ENQUEUE (CQ, 120); (viii) DEQUEUE (CQ); (ix) DEQUEUE (CQ); (X) ENQUEUE (CQ, 10);

Answers are expected as per the format given in Table 2;

| Command | Circular queue | Front | Rear |
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5. Differentiate iterative procedure and recursive procedure. Write the recursive algorithm for finding the power of a given number and trace the algorithm for power values of (0, 1, odd, even). Derive the recurrence relation for the power function and analyze its time complexity.

**Description of each step should be given. Verification of the answers should be highlighted**

**Note: Assignment should be written in A4 paper with proper highlights for all the answer and diagrams should be drawn using pencil.**